A PROJECTIVE COVER, WHEN IT EXISTS, IS UNIQUE UP TO ISOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The defining properties of a projective cover: the map is essential and the source is projective. Used throughout the counted proof (source lines 645-652). Supplies projectivity and essentialness of the two covers.

Lemma 0.1. *Let R be a ring and let M be an R-module. If a projective cover of M exists, then it is unique up to isomorphism.*

Proof. Let $P \to M$ and $P' \to M$ be projective covers. Because P is a projective R-module and $P' \to M$ is surjective, we can find an R-module map $\alpha : P \to P'$ compatible with the maps to M. Since $P' \to M$ is essential, we see that α is surjective. As P' is a projective R-module we can choose a direct sum decomposition $P = \mathrm{Ker}(\alpha) \oplus P'$. Since $P' \to M$ is surjective and since $P \to M$ is essential we conclude that $\mathrm{Ker}(\alpha)$ is zero as desired. □

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